Probability Tree Diagrams

Exam code: 0607

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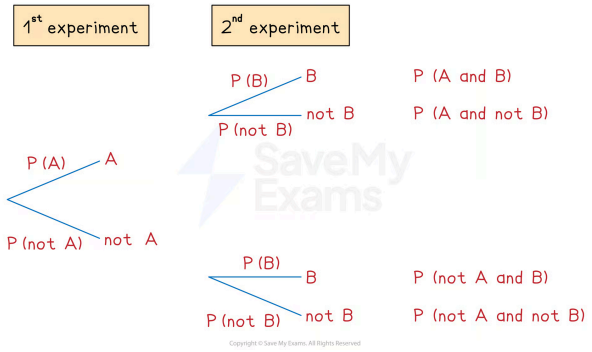
Updated on 23 January 2026

Tree diagrams

How do I draw a tree diagram?

- **Tree diagrams** can be used for **repeated experiments** with **two outcomes**
  - The **1st experiment** has outcome **A** or **not A**
  - The **2nd experiment** has outcome **B** or not **B**
- Read the tree diagram from **left to right** along its **branches**
  - For example, the top branches give A followed by B
    - This is called A **and** B



How do I find probabilities from tree diagrams?

- Write the **probabilities** on each branch
  - Remember that P(**not** A) = 1 - P(A)
    - Probabilities on each **pair** of branches **add to 1**
- **Multiply** along the branches from **left to right**
  - This gives P(1st outcome **and** 2nd outcome)
- **Add** between the **separate cases**
  - For example
    - P(AA **or** BB) = P(AA) + P(BB)
- The probabilities of **all** possible cases **add to 1**
- If asked to find the probability of **at least one** outcome, it is quicker to do **1 – P(none)**

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Examiner Tips and Tricks

When multiplying along branches with fractions, don't cancel fractions in your working - having the same denominator makes them easier to add together!

Worked Example

A worker drives through two sets of traffic lights on their way to work.
Each set of traffic lights has only two options: green or red.
The probability of the first set of traffic lights being on green is $\frac{5}{7}$.
The probability of the second set of traffic lights being on green is $\frac{8}{9}$.

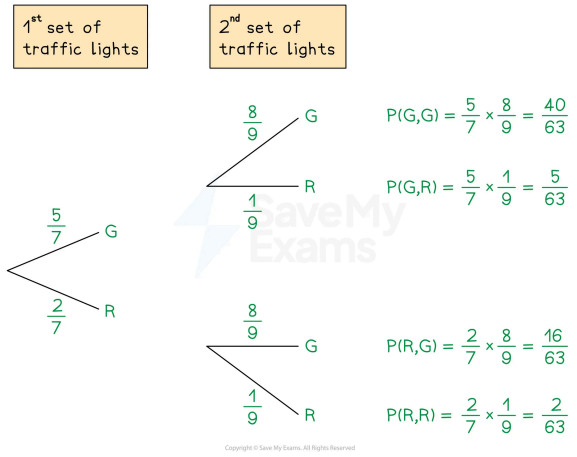
(a) Draw and label a tree diagram. Show the probabilities of every possible outcome.

Answer:

Work out the probabilities of each set of traffic lights being on red, R
Use P(red) = 1 - P(green)

$$P(1^{st}\ R) = 1 - P(1^{st}\ G) = 1 - \frac{5}{7} = \frac{2}{7}$$
$$P(2^{nd}\ R) = 1 - P(2^{nd}\ G) = 1 - \frac{8}{9} = \frac{1}{9}$$

Draw the branches (with a label of G or R on the ends)
Write the probabilities above each branch
Calculate probabilities of each outcome by multiplying along the branches from left to right



(b) Find the probability that both sets of traffic lights are on red.

Answer:

$\frac{}{63}$

(c) Find the probability that at least one set of traffic lights is on red.

Answer:

Method 1

This means the 1st is green and the 2nd is red
Or the 1st is red and the 2nd is green
Or the 1st is red and the 2nd is red ('at least one' could mean both)
Add between the separate cases

$$P(\text{at least one R}) = P(G, R) + P(R, G) + P(R, R) = \frac{5}{63} + \frac{16}{63} + \frac{23}{63}$$

Method 2
At least one red means all the possible cases shown except two greens
So P(at least 1 red) = 1 - P(two greens)

$$1 - P(G, G) = 1 - \frac{40}{63} = \frac{23}{63}$$

$$\frac{23}{63}$$

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